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Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 1

§1.2 Basic Concepts

Tuesday 8 September 2026

Where to find course information

- The public course web site is [Math 2C03: https://davidearn.github.io/math2c/](https://davidearn.github.io/math2c/)
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 lecture recordings and course announcements.
- *You are expected to read and pay attention to every word of the course syllabus.*
- Let's have a look now...
- All assignments will be on [childsmath](https://www.childsmath.ca/childsa/forms/main_login.php): https://www.childsmath.ca/childsa/forms/main_login.php

Things to review from previous courses

- Chain rule
- Ordinary derivatives, partial derivatives, implicit differentiation
- Change of variables
- Methods of integration including: partial fraction decomposition, integration by parts, trig substitution
- Simple separable and linear differential equations
- Convergence of infinite sequences and series
- Linear algebra: solving systems of linear equations, linear independence, basis

Refresher resources on [childsmath](#):

- Pre-Calculus Review
- Calculus Warm-up

Poll to check you're awake: Identify the instructor

- Please log in (right now) to the [childsmath](https://www.childsmath.ca/childsa/forms/main_login.php) web site: https://www.childsmath.ca/childsa/forms/main_login.php
- Click on [Math 2C03](#).
- Click on [Take Class Poll](#).
- After selecting the person you think is your instructor, click the button.
- Everybody done?
- Let's deactivate the poll and view the results.

What is a differential equation?

- First, consider (simpler) examples of *algebraic equations*:
 - $2x + 7 = 0$;
 - $x^2 + x - 1 = 0$;
 - $x^2 + 1 = 0$.
- In each of the above cases, x is an *unknown real number*.
- Such an algebraic equation can have:
 - one solution
 - a *unique* real number x satisfies the equation;
 - more than one solution
 - multiple real values of x satisfy the equation;
 - no solutions
 - no $x \in \mathbb{R}$ satisfies the equation.

What is a differential equation?

- Next, consider an *algebraic equation* involving two variables,

$$x^2 + y^2 = 1 \quad (1)$$

- We can ask: what points $(x, y) \in \mathbb{R}^2$ satisfy this equation?
 - Answer: all the points on the unit circle (and no others).
- Or, we can think of x as *independent variable*, and y as a *dependent variable*, and ask: is there a function ϕ , such that if $y = \phi(x)$ then Equation (1) is satisfied for every x in some interval?
 - Answer: YES, there are two such functions:

$$\phi_1(x) = \sqrt{1 - x^2} \quad \text{for } -1 \leq x \leq 1,$$

$$\phi_2(x) = -\sqrt{1 - x^2} \quad \text{for } -1 \leq x \leq 1.$$

- $y = \phi_1(x)$ and $y = \phi_2(x)$ are *solutions* of Equation (1).

What is a differential equation?

- An equation for an *unknown function* is a *differential equation (DE)* if it involves derivatives of the unknown function.
 - Equivalently, an equation is a *differential equation* if it involves derivatives of a dependent variable, for which the functional relationship between the dependent and independent variable is unknown.
- A function that satisfies a DE on an interval is a *solution* of the DE.
 - Finding such a solution means discovering a function $\phi(x)$ such that setting $y = \phi(x)$, and simplifying, reduces the equation to an identity, e.g., " $0 = 0$ ".

What is a differential equation?

Example (Differential Equation)

$$\frac{dy}{dx} + x + y = 0 \quad (2)$$

Does this equation have a solution? *i.e.*, Is there a function $y = \phi(x)$ that satisfies Equation (2) on some interval?

Answer: YES, $y = \phi(x) = ce^{-x} - x + 1$ is a solution of Equation (2) on all of $\mathbb{R}$ for any $c \in \mathbb{R}$.

$$\begin{aligned} \text{Check: } \frac{dy}{dx} &= \frac{d}{dx} \phi(x) = \frac{d}{dx} (ce^{-x} - x + 1) = -ce^{-x} - 1 \\ \frac{dy}{dx} + x + y &= (-ce^{-x} - 1) + x + (ce^{-x} - x + 1) = 0 \end{aligned}$$

Note: if we specify a value for c , we obtain a **particular solution**. One particular solution is $y = 2e^{-x} - x + 1$.

Implicit solutions

Solutions of DEs are sometimes expressed *implicitly*, i.e., by an algebraic relation, say $G(x, y) = 0$.

To check such solutions, use implicit differentiation.

Example (Differential Equation with implicit solution)

$$\frac{dy}{dx} = -\frac{x^3}{y^3} \quad (3)$$

Proposed implicit solution: $G(x, y) = x^4 + y^4 - 1 = 0$.

Check:

$$0 = \frac{d}{dx}(x^4 + y^4 - 1) = 4x^3 + 4y^3 \frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{x^3}{y^3}$$

Definition of a differential equation

- In general, differential equations can involve ordinary derivatives and/or partial derivatives of one or more unknown functions of one or more variables.

Definition (Differential equation)

A *differential equation (DE)* is an equation that contains one or more derivatives of an unknown function.

- The *order* of a DE is the order of the highest order derivative in the DE.

Types of differential equations

- If a DE involves only ordinary derivatives, we call it an **ordinary differential equation (ODE)**, e.g.,

$$\frac{dy}{dx} + y = 0$$

independent variable: x

dependent variable: y

order: 1

$$2\frac{d^2x}{dt^2} + t^2\frac{dx}{dt} + (1 + e^t)x + 5 = 0$$

independent variable: t

dependent variable: x

order: 2

- If a DE involves partial derivatives, we call it a **partial differential equation (PDE)**, e.g.,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{1}{\sqrt{1 + x^2 + y^2}}$$

independent variables: x, y

dependent variable: u

order: 2

General vs Normal form of ODEs

The general n^{th} order ODE in **general form** is

$$F(x, y, y', y'', \dots, y^{(n)}) = 0, \quad (4)$$

where prime denotes derivative with respect to x .

In Equation (4), if it is possible to solve for the highest derivative, $y^{(n)}$, then we can re-write the equation in **normal form**,

$$y^{(n)} = f(x, y, y', y'', \dots, y^{(n-1)}). \quad (5)$$

Example (General $\rightarrow$ Normal form)

$$e^x \frac{d^2 y}{dx^2} - y^2 \frac{dy}{dx} + x^2 = 0 \quad \longrightarrow \quad \frac{d^2 y}{dx^2} = e^{-x} y^2 \frac{dy}{dx} - x^2 e^{-x}$$

Dynamical Systems

DEs that involve time as an independent variable are examples of *dynamical systems*.

Example (Newton's Second Law)

$$F(t, x, \dot{x}) = m\ddot{x}, \quad (6)$$

where

$$\text{velocity} \quad v = \dot{x} = \frac{dx}{dt}$$

$$\text{acceleration} \quad a = \ddot{x} = \frac{d^2x}{dt^2}$$

A solution of Equation (6), $x(t)$, is called a *trajectory*.

Applications of the theory of dynamical systems

In my research, I have applied the theory of dynamical systems to problems in physics and biology.

- astrophysics (solar systems, galaxies)
- evolution (game theory, replicators, stochastic processes)
- ecology (population dynamics, conservation)
- epidemiology (infectious disease transmission dynamics)

Curious about how I've used DEs and other tools in my research?

- check out the publications page on my web site:
<https://davidearn.mcmaster.ca/>
- or ask me...

Predicting epidemics with ODEs

Example (SIR equations: An ODE model of an epidemic)

Divide population into compartments:



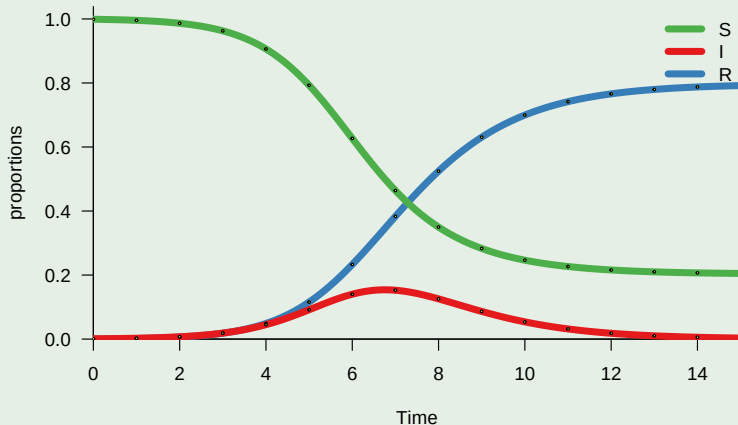
Express idea with a *system of three ODEs*:

$$\frac{dS}{dt} = -\beta SI, \quad \frac{dI}{dt} = \beta SI - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

Predicting epidemics with ODEs

Example (SIR equations: An ODE model of an epidemic)

Solution is a set of three functions: $S(t)$, $I(t)$, $R(t)$



Predicting epidemics with ODEs

- even undergrads can contribute...

INTERFACE

royalsocietypublishing.org/journal/rsif

Research



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Face masking and COVID-19: potential effects of variolation on transmission dynamics

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Face masks do not completely prevent transmission of respiratory infections, but masked individuals are likely to inhale fewer infectious particles. If smaller infectious doses tend to yield milder infections, yet ultimately induce similar levels of immunity, then masking could reduce the prevalence of severe disease even if the total number of infections is unaffected. It has been suggested that this effect of masking is analogous to the pre-vaccination practice of variolation for smallpox, whereby susceptible individuals were intentionally infected with small doses of live virus (and often acquired immunity without severe disease). We present a simple epidemiological model in which mask-induced variolation causes milder infections, potentially with lower transmission rate and/or different dur-

Predicting epidemics with ODEs

- more complicated than the SIR model...

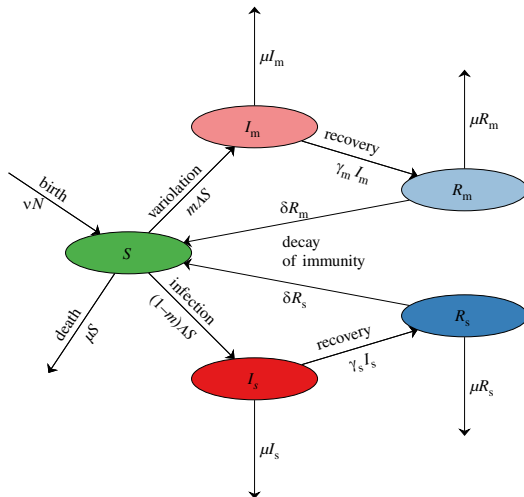


Figure 1. Flow chart for the model defined by equation (3.1). The state variables are defined in table 1, and the parameters are defined in table 3.

Predicting epidemics with ODEs

- but still represented with ODEs...

$$\frac{dS}{dt} = \nu N - \Lambda S - \mu S + \delta(N - S - I_m - I_s), \quad (3.1a)$$

$$\frac{dI_m}{dt} = m\Lambda S - \gamma_m I_m - \mu I_m, \quad (3.1b)$$

$$\frac{dI_s}{dt} = (1 - m)\Lambda S - \gamma_s I_s - \mu I_s, \quad (3.1c)$$

$$\frac{dR_m}{dt} = \gamma_m I_m - \mu R_m - \delta R_m \quad (3.1d)$$

$$\text{and } \frac{dR_s}{dt} = \gamma_s I_s - \mu R_s - \delta R_s, \quad (3.1e)$$

where the *force of infection* is

$$\Lambda = \beta_m \frac{I_m}{N} + \beta_s \frac{I_s}{N}, \quad (3.1f)$$

and the total population size is

$$N = S + I_m + I_s + R_m + R_s. \quad (3.1g)$$

Improving understanding

- Read §1.2 of the textbook.
- Attempt Exercises 1 and 2 at the end of §1.2.



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Mathematics 2C03

Introduction to Differential Equations

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Lecture 2

§1.2 IVPs; §2.3 Existence and Uniqueness

Thursday 10 September 2026

Announcements

- The public course web site is [Math 2C03: https://davidearn.github.io/math2c/](https://davidearn.github.io/math2c/)
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 lecture recordings and course announcements.
- Assignments and class polls are administered through [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, immediately after class.

More terminology – Classification of ODEs by Linearity

If V and W are vector spaces and $L : V \rightarrow W$, then L is **linear** if

$$L[au + bv] = aL[u] + bL[v]$$

for any $u, v \in V$ and any $a, b \in \mathbb{R}$.

If L is not linear, we say it is **nonlinear**.

For an ODE, V and W are spaces of functions, and a **linear differential operator** has the form

$$L[y] = a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y.$$

An n^{th} -order ODE is **linear** if it can be written

$$L[y] = g(x),$$

where L is a linear differential operator.

More terminology – Classification of ODEs by Linearity

Example (Linear vs nonlinear ODEs)

$$x^2 \frac{d^2 y}{dx^2} + e^x y - 3 \sin x = 0 \quad \text{linear}$$

$$x^2 \frac{d^2 y}{dx^2} + e^y y - 3 \sin x = 0 \quad \text{nonlinear}$$

$$x^2 \frac{d^2 y}{dx^2} + e^x y \frac{dy}{dx} - 3 \sin x = 0 \quad \text{nonlinear}$$

$$x^2 \frac{d^2 y}{dx^2} + e^x \frac{dy}{dx} - 3 \sin x = 0 \quad \text{linear}$$

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Understanding linear ODEs**
- .

Initial-Value Problems (IVPs)

In an *initial-value problem (IVP)*, we seek a solution $y(t)$ of a DE that satisfies *initial conditions* at an initial time t_0 .

The generic n^{th} -order IVP is:

Solve:

$$\frac{d^n y}{dt^n} = f(t, y, \dot{y}, \ddot{y}, \dots, y^{(n-1)}),$$

subject to initial conditions:

$$y(t_0) = y_0, \quad \dot{y}(t_0) = y_1, \quad \dots, \quad y^{(n-1)}(t_0) = y_{n-1}$$

Initial-Value Problems (IVPs)

Example (object of mass m thrown straight up with velocity v_0)

$$\begin{array}{ll} \text{Solve} & m \ddot{y} = -m g, \\ \text{subject to} & y(0) = 0, \dot{y}(0) = v_0 \end{array} \quad (7)$$

Solution of this IVP: $y(t) = v_0 t - \frac{1}{2} g t^2.$

Check:

$$y(0) = 0$$

$$\dot{y}(0) = \left. \frac{d}{dt} \left(v_0 t - \frac{1}{2} g t^2 \right) \right|_{t=0} = \left. (v_0 - g t) \right|_{t=0} = v_0$$

Question: Is the solution $y(t)$ valid for all time $t \geq 0$?

Answer: Yes, mathematically. No, physically!

Initial-Value Problems (IVPs)

If the independent variable is not time, we still refer to “initial-value” problems. The following is identical to the [slide before last](#), except for the notation:

In an ***initial-value problem (IVP)***, we seek a solution $y(x)$ of a DE that satisfies ***initial conditions*** at x_0 .

The generic n^{th} -order IVP is:

Solve:

$$\frac{d^n y}{dx^n} = f(x, y, y', y'', \dots, y^{(n-1)}),$$

subject to initial conditions:

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1}$$

Existence and Uniqueness

When solving an IVP, we need to determine if a solution *exists* and, if so, whether the solution is *unique*.

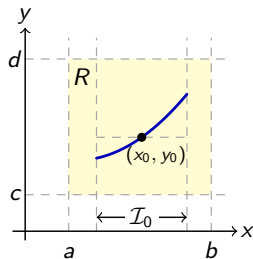
Existence: Does the DE possess solutions and, if so, do any solution curves pass through the point (x_0, y_0) ?

Uniqueness: Is there precisely one solution curve passing through the point (x_0, y_0) ?

Existence and Uniqueness

Trench Theorem 2.3.1 gives *sufficient* conditions that guarantee existence and uniqueness of a solution $y(x)$ to a *first-order* IVP:

$$\text{Solve: } \frac{dy}{dx} = f(x, y), \quad \text{subject to: } y(x_0) = y_0. \quad (8)$$



- rectangular region $R = [a, b] \times [c, d]$
- point $(x_0, y_0) \in R$ (not on the boundary)
- $f(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ continuous in R

*These conditions imply there exists an interval $\mathcal{I}_0 \subset (a, b)$, with $x_0 \in \mathcal{I}_0$, such that IVP (8) has a *unique solution* on $\mathcal{I}_0$.*

Theorem (Trench Theorem 2.3.1)

Let $R = \{a < x < b, c < y < d\}$ be an open rectangle that contains (x_0, y_0) .

- (a) If f is continuous on R , then IVP (8) has at least one solution on some open subinterval of (a, b) that contains x_0 .
- (b) If both f and f_y are continuous on R , then IVP (8) has a unique solution on some open subinterval of (a, b) that contains x_0 .

Existence and Uniqueness

Example (Does this IVP possess a unique solution?)

Solve: $\frac{dy}{dx} = y\sqrt{1+|x|},$ *subject to:* $y(0) = 1$

Example (Does this IVP possess a unique solution?)

Solve: $\frac{dy}{dx} = \frac{10}{3}xy^{2/5},$ *subject to:* $y(0) = 0$

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
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- Click on Take Class Poll
- Fill in poll **Existence and uniqueness**
- .

Existence and Uniqueness

Example (Does this IVP possess a unique solution?)

Solve: $\frac{dy}{dx} = y\sqrt{1+|x|}$, *subject to:* $y(0) = 1$

Answer: Yes. Consider the hypotheses of the [theorem](#).

$x_0 = 0, y_0 = 1$ i.e., “initial point” is $(x_0, y_0) = (0, 1)$

$f(x, y) = y\sqrt{1+|x|}$ continuous on all of $\mathbb{R}^2$, hence continuous on any rectangle R containing (x_0, y_0) .

$\frac{\partial f}{\partial y}(x, y) = \sqrt{1+|x|}$ continuous on all of $\mathbb{R}^2$, hence continuous on any rectangle R containing (x_0, y_0) .

$\therefore$ [Theorem](#) implies unique solution on some interval containing x_0 .

Existence and Uniqueness

Example (Does this IVP possess a unique solution?)

Solve: $\frac{dy}{dx} = f(x, y) = \frac{10}{3}xy^{2/5}$, *subject to:* $y(0) = 0$

Answer: Maybe. Consider the hypotheses of the [theorem](#).

$x_0 = 0$, $y_0 = 0$ i.e., “initial point” is $(x_0, y_0) = (0, 0)$

$f(x, y) = \frac{10}{3}xy^{2/5}$ continuous on all of $\mathbb{R}^2$, hence continuous on any rectangle R containing $(x_0, y_0) = (0, 0)$.

$\frac{\partial f}{\partial y}(x, y) = \frac{4}{3}xy^{-3/5}$ not even defined for $y = 0$ (x -axis)
 $\implies$ cannot be continuous on any rectangle R that contains $(x_0, y_0) = (0, 0)$.

$\therefore$ [Theorem](#) guarantees existence but does not imply uniqueness!

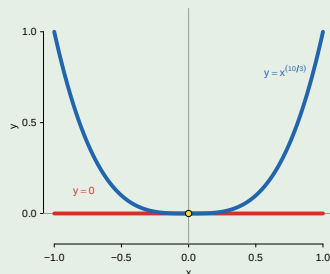
Existence and Uniqueness

Example (Does this IVP possess a unique solution?)

Solve: $\frac{dy}{dx} = f(x, y) = \frac{10}{3}xy^{2/5},$ *subject to:* $y(0) = 0$

Solution curves:

Both $y(x) = 0,$
and $y(x) = x^{10/3}$
satisfy the IVP.



Existence and Uniqueness

Example (Does this IVP possess a unique solution?)

Solve: $\frac{dy}{dx} = f(x, y) = \frac{10}{3}xy^{2/5},$ *subject to:* $y(0) = 0$

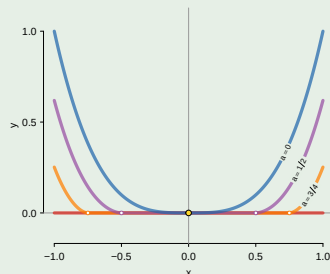
Solution curves:

In fact, this IVP has *a continuum of solutions*.

For every $a \geq 0$,

$$y_a(x) = \begin{cases} 0, & |x| \leq a, \\ (x^2 - a^2)^{5/3}, & |x| > a, \end{cases}$$

Each y_a solves the IVP.



Existence and Uniqueness

Enrichment (not tested): How many solutions can an IVP have?

Theorem (Kneser's theorem (simplified form))

Suppose f is continuous on a rectangle containing (x_0, y_0) . On a sufficiently short interval $[x_0, x_0 + h]$, the IVP

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0,$$

has either *exactly one solution* or *a continuum of solutions*.

"A continuum" means as many as there are real numbers in $[0, 1]$. Thus, the number of solutions to the IVP cannot be exactly 2, 3, or any other finite $n > 1$, nor can there be merely an infinite list $y_1, y_2, y_3, \dots$

The preceding example illustrates the continuum possibility: every $a \geq 0$ gives a different solution y_a .

Important: Failure of the derivative-based uniqueness test does not by itself imply nonuniqueness.

References: Wikipedia: H. Kneser's theorem

P. Hartman, *Ordinary Differential Equations*, Chapter II, §4, Theorem 4.1, p. 15.